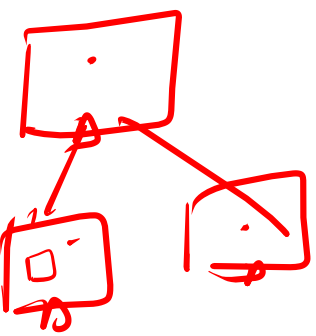
```
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP               NODE                NO
mydep-54c57f8d57-pwmrs      1/1     Running   0          23h   192.168.9.151    ip-172-31-2-109     <n
mydep-54c57f8d57-wcfsc      1/1     Running   0          23h   192.168.122.87   ip-172-31-15-78     <n
root@ip-172-31-6-141:/home/ubuntu# kubectl delete pods --all
pod "mydep-54c57f8d57-pwmrs" deleted
pod "mydep-54c57f8d57-wcfsc" deleted
root@ip-172-31-6-141:/home/ubuntu# kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP               NODE
mydep-54c57f8d57-lj2f9      1/1     Running   0          2m24s   192.168.9.152    ip-172-31-2-109
mydep-54c57f8d57-q6qps      1/1     Running   0          2m24s   192.168.122.88   ip-172-31-15-78
root@ip-172-31-6-141:/home/ubuntu#
```

Changed

if due to any reason pods are restarted then the ip address of the pods will be changed ...and if the pod is interactng with some endpoint then that connection string will break. to avoid it we should be able to provide a static ip and this concept is service in k8s

Service → Pods (your containers) comes and go.
The ip address changes.
A service is a way to reach the pods - like a name or number that doesnt change by providing the static ip.



Cluster IP

like a fixed private ip

Static ip but the ip is only accsible from within the Cluster.

Node port

Expose the service on each Node's IP at a static port (30000-32767) & content of pod is accsible from Internet.